

Combination

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$$v = u - gt \quad \text{and} \quad s = \left(\frac{u + v}{2} \right) t$$

Comments: From definitions.

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$$s = \left(\frac{u + (u - gt)}{2} \right) t$$

Comments: Distribute the terms.

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$$s = \left(\frac{u + (u - gt)}{2} \right) t \quad \Rightarrow \quad s = ut - \frac{1}{2}gt^2$$

Comments: That's it.